

Module

3

Operating System Fundamentals - Theoretical Content



Linux Fundamentals



Linux overview

- Open Source Operating System
 - Kernel developed by Linus Torvalds (1991).
 - Available in many different distributions (distros).
 - Dominant in server and cloud environments.
 - Heavily focused on the command-line interface.

Linux Distributions

Popular distributions:

- Ubuntu - User-friendly, enterprise
- Debian - Stable, conservative
- Red Hat / CentOS - Enterprise
- Kali Linux - Security testing
- Arch - Minimal, advanced

Focus: Ubuntu (learning), Kali (security)

Linux Architecture

- Hardware: The physical components of the system.
- Kernel: The core of the operating system, managing resources.
- Shell: The command interpreter, providing an interface for the user.
- System utilities: Programs that perform specific tasks for system administration or maintenance.
- Applications: Software programs used by the end-user.

Linux File System Hierarchy

Key directories:

- / - Root (top level)
- /home - User directories
- /etc - Configuration files
- /var - Variable data (logs)
- /bin - Essential binaries
- /usr - User programs
- /tmp - Temporary files

Linux File System Types

- Common types:
 - ext4 - Default, reliable
 - XFS - High performance
 - Btrfs - Modern, snapshots
 - ZFS - Advanced features
- We use: ext4 (standard)

Linux Permissions - Critical Concept

Three permission types:

- r (read) - View file
- w (write) - Modify file
- x (execute) - Run file

Three user categories:

- Owner - File creator
- Group - Shared access
- Others - Everyone else

Reading Linux Permissions

-rwxr-xr-- 1 user group 1234 Oct 20 file.txt

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| | | └─ Others: r-- (read only)

| | └─── Group: r-x (read, execute)

| └────── Owner: rwx (all)

└──────── Type: - (file)

Numeric Permissions (chmod)

- Octal representation:
 - $r = 4$
 - $w = 2$
 - $x = 1$
- Examples:
 - $755 = \text{rwxr-xr-x}$ (common)
 - $644 = \text{rw-r--r--}$ (files)
 - $600 = \text{rw-----}$ (private)
 - $777 = \text{rwxrwxrwx}$ (dangerous!)

Special Permissions

- Advanced permissions:
 - SUID (4000) - Run as file owner
 - SGID (2000) - Run as file group
 - Sticky bit (1000) - Delete protection
- Security implications!

Linux Users and Groups

- User system:
 - root - Superuser (UID 0)
 - Regular users (UID 1000+)
 - System users (UID 1-999)
- Groups provide shared access

sudo - Temporary Elevation

- Safer than root login:
 - Executes single command as root
 - Requires user's password
 - Logged in /var/log/auth.log
 - Configurable per user/command
- Example: `sudo apt update`

Linux Shell - Command Interface

- Common shells:
 - bash - Bourne Again Shell (default)
 - zsh - Extended bash
 - sh - Original shell
 - fish - User-friendly
- We use: bash

Linux Package Management

- Package managers:
 - apt (Debian/Ubuntu)
 - yum/dnf (Red Hat/CentOS)
 - pacman (Arch)
- Install software from repositories

Linux System Logging

- Key log files:
 - /var/log/syslog - System messages
 - /var/log/auth.log - Authentication
 - /var/log/kern.log - Kernel messages
 - /var/log/apache2/ - Web server
- Plain text files!

Systemd - Service Management

- Modern init system:
 - Manages services and processes
 - Start/stop/restart services
 - Enable/disable autostart
 - View status and logs
- Command: systemctl

Part 4: Security Hardening



What is System Hardening?

- Reducing attack surface:
 - Disable unnecessary services
 - Apply security patches
 - Configure secure settings
 - Remove unused software
 - Make system resistant to attacks

Windows Hardening Principles

- Enable Windows Update (automatic)
- Configure Windows Firewall
- Enable BitLocker encryption
- Disable unnecessary services
- Apply Group Policy security
- Use standard user accounts
- Install only needed software

Linux Hardening Principles

-  Keep system updated
-  Configure firewall (ufw/iptables)
-  Disable root login
-  Use SSH keys (not passwords)
-  Remove unused packages
-  Configure SELinux/AppArmor
-  Restrict file permissions

Patch Management

- Critical for security:
 - Patches fix known vulnerabilities
 - Attackers exploit unpatched systems
 - Test patches before deployment
 - Automate when possible
- Windows: Windows Update
- Linux: apt/yum update

User Account Security

Best practices:

- Enforce strong passwords
- Implement password expiration
- Require password history
- Lock accounts after failed attempts
- Disable unused accounts

Firewall Configuration

- Defense Layer:
 - Block inbound by default
 - Allow only necessary services
 - Restrict by IP when possible
 - Log denied connections
 - Review rules regularly

Least Privilege Principle

Give minimum needed access:

- Users: Standard accounts daily
- Admins: Elevate only when needed
- Services: Dedicated accounts
- Applications: Minimal permissions

Why: Limit damage from compromise

Logging and Monitoring

- You can't protect what you can't see:
 - Enable comprehensive logging
 - Monitor critical events
 - Set up alerts
 - Regular log review
 - Logs = incident response foundation

Backup and Recovery - Ransomware Defense

- Backup strategy (3-2-1 Rule):
 - 3 copies of data.
 - 2 different media types.
 - 1 offsite/offline copy.
- Test restores regularly!

Security Baselines

- Standard Secure Configuration:
 - Document secure settings
 - Apply to all systems consistently
 - CIS Benchmarks (industry standard)
 - Automated compliance checking
 - Update baselines regularly

Thank you!!